School of Informatics & Cybersecurity - Assessment Brief Coversheet

Assessment Information

Module Title:	**Network Security**
Module Code:	**DFCS H3015**
Assessment Title:	**CA1 - Open Source Firewalls Assignment**
Assessment Type (e.g. Essay, Report, Presentation, Code submission):	**Report, Presentation & Demonstration**
Assessment Weighting (%):	**35% CA**
Submission Deadline:	**11.59pm 28/02/2026**
Submission mechanism:	**Report upload to Brightspace, Presentation & Demonstration in the lab**
Late Submission Penalty:	**0% for late submissions**

Use of AI

1	**NO AI**	The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their own knowledge, understanding, and skills. **AI Must not be used at any point during the assessment.**	☐
2	**AI-ASSISTED IDEA GENERATION AND STRUCTURING**	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. **No AI content is allowed in the final submission.**	☑
3	**AI-ASSISTED EDITING**	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. **AI can be used, but your original work with no AI content must be provided in an appendix.**	☐
4	**AI TASK COMPLETION, HUMAN EVALUATION**	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI-generated content and evaluating its output. **You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.**	☐
5	**FULL AI**	AI should be used as a 'co-pilot' to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. **You may use AI throughout your assessment to support your own work and do not have to specify which content is AI-generated.**	☐

Further conditions for the use of AI not covered above:

This is covered above but just to reiterate that AI tools are NOT permitted for this assessment to generate tests or firewall rules, nor can it be used for your background research.

Lecturer name:	Peter Alexander
Date:	30/01/2026

CA1 – Open Source Firewalls Assignment Specification

Course: BSc. in Computing in Digital Forensics and Cyber Security (TU758/TU863)

Module Title: Network Security **Module Code: DFCS H3015**

Lecturer's Name: Peter Alexander **Email: peter.alexander@tudublin.ie**

Date Assignment given out: 30/01/26 **Latest date for Submission: 11.59pm 28/02/26**

% of total CA marks allocated: 35% (of CA) **Group Size: 2-3 (must be done in groups of minimum 2)**

General Requirements for Students:

1. A proportion of assessment marks is allocated to the report presentation. All assignment scripts must be word-processed, where appropriate.
2. The **School of Informatics & Cybersecurity - Assessment Submission Coversheet** must be included with each assignment prior to submission.
3. Assignment scripts must be submitted via link on Brightspace. Email submissions will not be accepted.
4. All relevant provisions of the Assessment Regulations must be complied with. Penalties for late submission of assignments are as follows:

   No marks for assignments submitted after the deadline.

5. Extensions to assignment submission deadlines will only be granted in exceptional circumstances. The appropriate "Application for Extension" form must be used and supporting documentation (e.g. medical certificate) must be attached. Applications for extensions should be made directly to the Programme Leader **in advance** of the deadline date.
6. Assignments that exceed the word count will be penalised.
7. Students are required to refer to the Assessment Regulations on the TU Dublin website.
8. TU Dublin penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying). Please refer to the referencing guidelines on Brightspace for information on correct referencing.
9. Please be aware that the use of such tools as LLMs (Large Language Models such as chatGPT) is unfair academic practice and, as listed in the QQI Academic Integrity Guidelines, will be treated as plagiarism. In summary, the work submitted for the assessment **must be your own work**.

Assignment Details

Outline of Assignment	Research and test two open source firewalls if working in a group

	of two or three open source firewalls if working in a group of three. One of the firewalls which must be Nftables. (*Note: One of the firewalls cannot be IPTables as this is being replaced by Nftables. UFW or Firewalld also will not be accepted as these are essentially frontend based versions of IPTables. Any other firewalls that are just a frontend for Nftables or IPTables are also not appropriate. If in doubt, please check with your lecturer.)* Implement and configure the 2-3 firewalls in a realistic testbed scenario. Design a series of tests to examine the effectiveness of the firewalls. Compare and contrast the 2-3 firewalls. Collate and analyse the results of your testing. Demonstrate your tests in the lab. Make a presentation on your findings (PowerPoint). Write a report on your assignment and findings (Maximum 2500 words for group of 2 or 3000 words for group of 3). Report should clearly demonstrate your implementation and test results using appropriate screenshots where necessary.
Detail of Assignment	Deliverables: • Report/paper. Maximum 2500 words in length (3000 if working in a group of 3). This should be written in an academic style, and fully referenced using Harvard style referencing. • Presentation (mandatory) in the lab: 5 minutes – strictly enforced. • Demonstration (mandatory) in the lab: 7 minutes – strictly enforced. ○ (*Note: This may be reduced to 5 minutes – I will confirm closer to the time.)* Submission is through a dedicated link on Brightspace. Late submissions will not be accepted. Files should be saved as MS Word or PDF files with the following filename structure: student number, name, netsecurity eg. B00012345WalterWhitenetsecurity.pdf
Lecturer this work is for	Peter Alexander
Handout date	30th January 2026
Hand-in date	Research paper to be submitted via Brightspace by 11.59pm on Saturday 28th February 2026. Presentations and demos will take place Monday & Tuesday (2nd & 3rd March 2026) during your assigned lab session. **IMPORTANT: The presentation and demo are mandatory requirements and any assignments without one will not be accepted.**
Group	This is a team project (maximum three persons per team).

Marking Scheme

Assessment Criteria	% weighting for each problem part
Appropriate background research.	10%
Effective configuration of firewall rules and demonstration of tests.	15%
Analysis of results presented in appropriate context.	15%
Extra relevant research/context.	10%
Proper references and overall quality of the report.	10%
Effective presentation with explanation of setup, overview of firewall configuration and showing an understanding of well-thought out, logical, relevant testing and analysis of results.	20%
Effective demonstration, showing understanding of firewall technologies and effective testing.	20%
(Total)	**100%**

*Note: The report, presentation and demonstration are each a mandatory component and any completed assignment that does not include all 3 components will be considered as an **invalid submission** which will receive **0%**.*

Other Information

What you should hand in	A structured report containing the research. Sources of information MUST be referenced. Harvard style referencing (see the resources on Brightspace).
Guidelines/Length	The report would be expected to be in the region of 2500 words (or 3000 words if in a group of 3), not including title page, table of contents and references.
Resource Required	Internet. Open source firewalls include Nftables, pfSense, OPNsense, Smoothwall, Endian, IPCop, IPFire etc.
Plagiarism	Your attention is drawn to TU Dublin's Academic Integrity guidelines, available on the website here. This covers cheating, attempts to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessments. The work you submit must conform to those regulations.

Common Pitfalls for this Assignment:

- **Poorly thought out tests**
- **Lack of diversity in tests**
    - **E.g. only testing using different Nmap scans – these are all different versions of the same test (i.e. port scans). Should be aiming for 2-3 different relevant tests**
- **Minimum/no configuration of firewall rules**
- **Poor understanding of firewall rules and/or poor demonstration of firewall capabilities**
- **Poorly thought out testbed topology/layout**
- **Inability to put tests in the correct context – why was each test carried out and why was it successfully/unsuccessfully mitigated**
- **Ineffective configuration of the firewalls – an important aspect of this assignment is to implement rules on the firewalls to mitigate attacks**
- **Lack of details on tests conducted**
- **Lack of effective comparisons**
- **Lack of detail in the analysis**
- **Weak conclusion based on analysis and data**
- **Failure to submit a report before deadline – always submit something rather than nothing**
- **Failure to attend presentation and/or demonstration**